

(Note: Complete all the question in your Machine Learning Notebook)

Confusion Matrix Numerical

Question 1:

A hospital is testing a new diagnostic tool to detect a specific disease. Out of 1,000 patients tested, 100 patients actually have the disease. The tool's predictions are as follows:

- 90 patients are correctly diagnosed with the disease.
- 150 patients are diagnosed with the disease but do not actually have it.
- 10 patients who have the disease are incorrectly diagnosed as not having it.
- 750 patients are correctly diagnosed as not having the disease.

Tasks:

- Create the confusion matrix.
- Calculate the following metrics:
 - Accuracy
 - Precision
 - Recall (Sensitivity)
 - Specificity
 - F1 Score

Question 2:

A credit card company is using a machine learning model to detect fraudulent transactions. The company tested the model on 10,000 transactions, where 500 transactions were fraudulent. The model made the following predictions:

- 450 transactions were correctly identified as fraudulent.
- 100 transactions were incorrectly identified as fraudulent.
- 50 fraudulent transactions were missed and classified as non-fraudulent.
- 9,400 transactions were correctly identified as non-fraudulent.

Tasks:

- Create the confusion matrix.
- Calculate the following metrics:
 - Accuracy
 - Precision
 - Recall (Sensitivity)
 - Specificity
 - F1 Score

Question 3:

A university is using a machine learning model to predict whether students will pass or fail a course based on their study habits and attendance. They tested the model on 200 students and obtained the following results:

- 160 students were correctly predicted to pass.
- 20 students were incorrectly predicted to pass but actually failed.
- 10 students were incorrectly predicted to fail but actually passed.
- 10 students were correctly predicted to fail.

Tasks:

- Create the confusion matrix.
- Calculate the following metrics:
 - Accuracy
 - Precision
 - Recall (Sensitivity)

- Specificity
- F1 Score

Question 4:

A company is testing a machine learning model to classify customer reviews as positive or negative. They tested the model on 1,500 reviews, where 1,000 were positive, and 500 were negative. The model produced the following results:

- 900 positive reviews were correctly identified.
- 50 positive reviews were incorrectly identified as negative.
- 100 negative reviews were incorrectly identified as positive.
- 400 negative reviews were correctly identified.

Tasks:

- Create the confusion matrix.
- Calculate the following metrics:
 - Accuracy
 - Precision
 - Recall (Sensitivity)
 - Specificity
 - F1 Score

Question 5:

A weather prediction model is used to forecast whether it will rain or not on a given day. Over the course of 300 days, the model made the following predictions:

- 180 days were correctly predicted as rainy.
- 30 days were predicted as rainy but were actually dry.
- 20 rainy days were missed and predicted as dry.
- 70 days were correctly predicted as dry.

Tasks:

- Create the confusion matrix.
- Calculate the following metrics:
 - Accuracy
 - Precision
 - Recall (Sensitivity)
 - Specificity
 - F1 Score